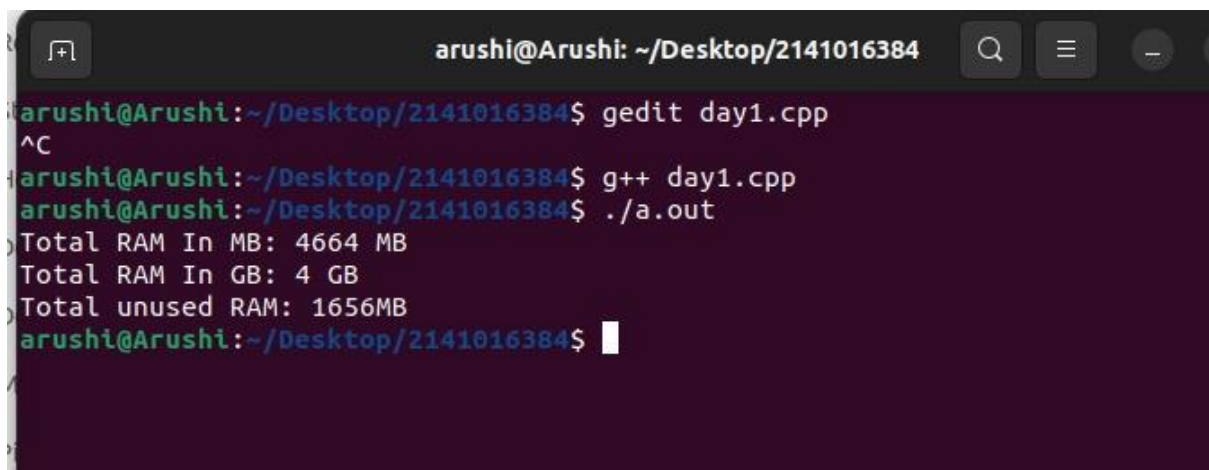


SYSTEM MONITOR TOOL

OBJECTIVE:-

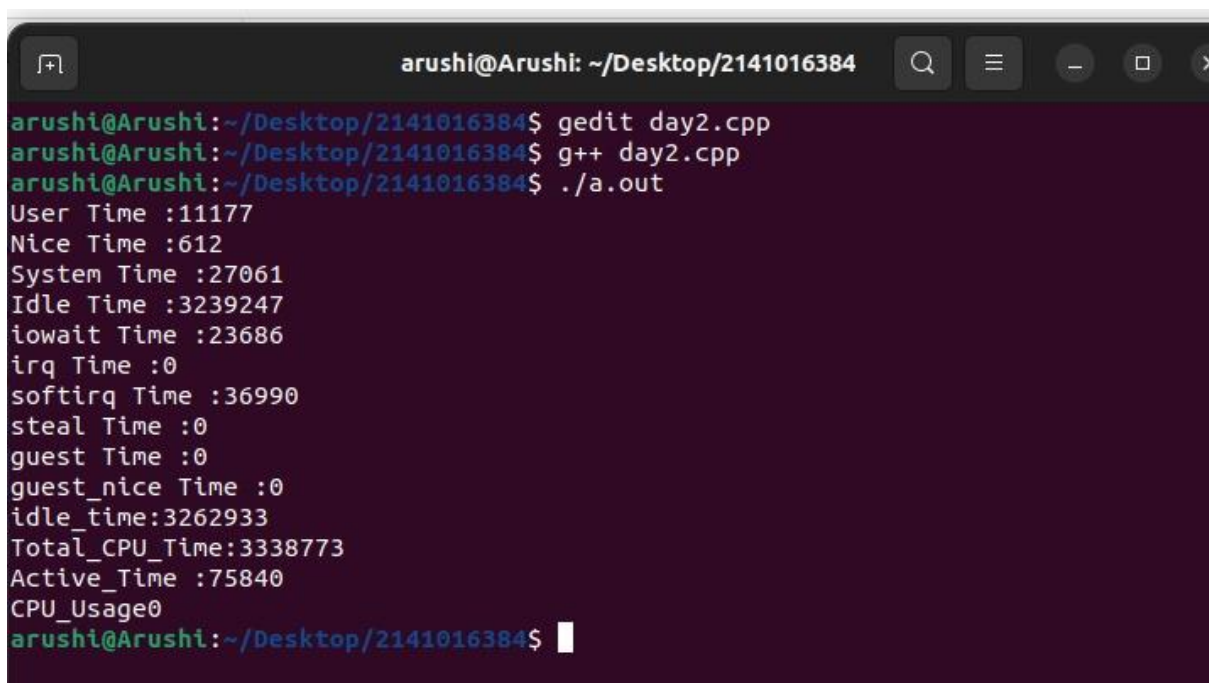
Create a system monitor tool in C++ that displays real time information about system processes, memory usage ,and CPU load ,similar to 'TOP' command.

DAY1: DESIGN UI LAYOUT AND GATHER SYSTEM DATA USING SYSTEM CALLS .

A terminal window with a dark purple background. The title bar shows 'arushi@Arushi: ~/Desktop/2141016384'. The terminal text shows the user running 'gedit day1.cpp', pressing Ctrl-C, then 'g++ day1.cpp', and finally './a.out'. The output displays system memory statistics: 'Total RAM In MB: 4664 MB', 'Total RAM In GB: 4 GB', and 'Total unused RAM: 1656MB'.

```
arushi@Arushi:~/Desktop/2141016384$ gedit day1.cpp
^C
arushi@Arushi:~/Desktop/2141016384$ g++ day1.cpp
arushi@Arushi:~/Desktop/2141016384$ ./a.out
Total RAM In MB: 4664 MB
Total RAM In GB: 4 GB
Total unused RAM: 1656MB
arushi@Arushi:~/Desktop/2141016384$
```

DAY2: DISPLAY PROCESS LIST WITH CPU AND MEMORY USAGE.

A terminal window with a dark purple background. The title bar shows 'arushi@Arushi: ~/Desktop/2141016384'. The terminal text shows the user running 'gedit day2.cpp', 'g++ day2.cpp', and './a.out'. The output displays various system time statistics: 'User Time :11177', 'Nice Time :612', 'System Time :27061', 'Idle Time :3239247', 'iowait Time :23686', 'irq Time :0', 'softirq Time :36990', 'steal Time :0', 'guest Time :0', 'guest_nice Time :0', 'idle_time:3262933', 'Total_CPU_Time:3338773', 'Active_Time :75840', and 'CPU_Usage0'.

```
arushi@Arushi:~/Desktop/2141016384$ gedit day2.cpp
arushi@Arushi:~/Desktop/2141016384$ g++ day2.cpp
arushi@Arushi:~/Desktop/2141016384$ ./a.out
User Time :11177
Nice Time :612
System Time :27061
Idle Time :3239247
iowait Time :23686
irq Time :0
softirq Time :36990
steal Time :0
guest Time :0
guest_nice Time :0
idle_time:3262933
Total_CPU_Time:3338773
Active_Time :75840
CPU_Usage0
arushi@Arushi:~/Desktop/2141016384$
```

DAY3: IMPLEMENT PROCESS SORTING BY CPU AND MEMORY USAGE :-

```
arushi@Arushi: ~/Desktop/2141016384
arushi@Arushi:~/Desktop/2141016384$ gedit day3.cpp
arushi@Arushi:~/Desktop/2141016384$ g++ day3.cpp
arushi@Arushi:~/Desktop/2141016384$ ./a.out
PID      CPU%    Memory (kB)    Name
2551     22.5659  0              (gnome-shell)
5646     11.3924  0              (a.out)
5332     1.48922  0              (nautilus)
1        1.31928  0              (systemd)
48       1.29149  0              (ksoftirqd/5)
754      1.24057  0              (dbus-daemon)
5641     1.05541  0              (tracker-extract)
765      0.99567  0              (polkitd)
2407     0.983548 0              (dbus-daemon)
755      0.923989 0              (NetworkManager)
arushi@Arushi:~/Desktop/2141016384$
```

DAY4: ADD FUNCTIONALITY TO KILL PROCESS:-

```
arushi@Arushi: ~/Desktop/2141016384
arushi@Arushi:~/Desktop/2141016384$ gedit day4.cpp
arushi@Arushi:~/Desktop/2141016384$ g++ day4.cpp
arushi@Arushi:~/Desktop/2141016384$ ./a.out
PID      CPU%    Memory (kB)    Name
8156     69.521  0              (gnome-shell)
8004     14.4099 0              (dbus-daemon)
8771     13.9154 0              (gnome-control-c)
7964     12.7775 0              (pulseaudio)
8862     8.8748  0              (nautilus)
8495     6.31791 0              (ibus-extension-)
8948     5.91065 0              (update-notifier)
8635     5.47807 0              (tracker-miner-f)
8994     5.45455 0              (a.out)
8114     4.94512 0              (gnome-session-b)
Enter PID to Kill :

```

DAYS5: IMPLEMENT REAL-TIME UPDATE FEATURE TO REFRESH DATA EVERY FEW SECONDS:-

```
arushi@Arushi: ~/Desktop/2141016384
arushi@Arushi:~/Desktop/2141016384$ gedit day5.cpp
^C
arushi@Arushi:~/Desktop/2141016384$ g++ day5.cpp
arushi@Arushi:~/Desktop/2141016384$ ./a.out
```

```
arushi@Arushi: ~/Desktop/2141016384
PID      CPU%    Memory (kB)    Name
9972     40.1591 0               (gnome-shell)
9818     6.48795 0               (dbus-daemon)
9780     4.89465 0               (pulseaudio)
10757    4.87805 0               (a.out)
10312    3.27763 0               (ibus-extension-)
10442    2.8309 0               (tracker-miner-f)
9932     2.62481 0               (gnome-session-b)
10187    2.45315 0               (nautilus)
10385    2.16463 0               (Xwayland)
765      2.11643 0               (polkitd)
Enter PID to Kill :
```

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